01. Business Problem & Objective

A small tabular multiclass classification problem with a highly imbalanced target.

Project: Child Malnutrition Status Classification (ICFDR ML Project)

Source archive: ICFDR---ML-Project-main.zip | Main notebook: MalnutritionDetect.ipynb | Additional notebooks: SMOTE.ipynb and Oversampling on the whole dataset.ipynb

Business context

The project uses child measurements and basic demographic/economic indicators to classify a record into one of four malnutrition-related status labels: Stunting, Overweight, Underweight or Wasting. The practical motivation is to support screening and prioritisation by identifying records that may need closer assessment.

ML formulation

This is supervised multiclass classification. Each row is one observation; the model learns a mapping from input features to one categorical target. It is not a regression task: the target labels represent categories rather than a continuous quantity.

Element	Project definition
Unit of observation	One child/record in the supplied tabular dataset
Features	Sex, Age, Height, Weight and three income-group indicator columns
Target	Status: Stunting, Overweight, Underweight, Wasting
Task	Four-class classification
Primary challenge	Severe class imbalance and very small sample size

Objectives

- Predict all four classes rather than defaulting to the majority Stunting class.
- Understand whether basic anthropometric and demographic features separate the labels.
- Compare baseline classifiers and imbalance-handling approaches.
- Evaluate minority-class recall and macro-F1, not accuracy alone.
- Keep the validation and test distributions representative of real-world data.

Important health-context qualification

This is an educational ML exercise, not a clinical diagnostic tool. A model trained on a small dataset cannot establish a child's nutritional status or replace age/sex-specific clinical growth standards and professional assessment. Any real use would require domain validation, representative data, careful governance and human oversight.

Interview framing

“I worked on a four-class classification problem using child-level demographic and anthropometric features such as age, height and weight. The main technical challenge was severe class imbalance: Stunting made up most observations, while Wasting had only four examples. I compared conventional classifiers and SMOTE-based training, and learned that accuracy alone can hide near-zero recall for rare classes.”